AP PHYSICS C

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0.1 Foreword for anybody else using this

As of writing this, I've already taken AP Physics 1 and 2 and Calc BC, so I'll skim over those stuff. I steal the images from google. If this violates copyright somehow, please email me and I'll remove it.

1 Unit 1: Kinematics

1.1 Scalars and Vectors

Scalars are numbers. Vectors are vectors. Represent as $\vec{a} = \hat{\imath} + \hat{\jmath} + \hat{k}$.

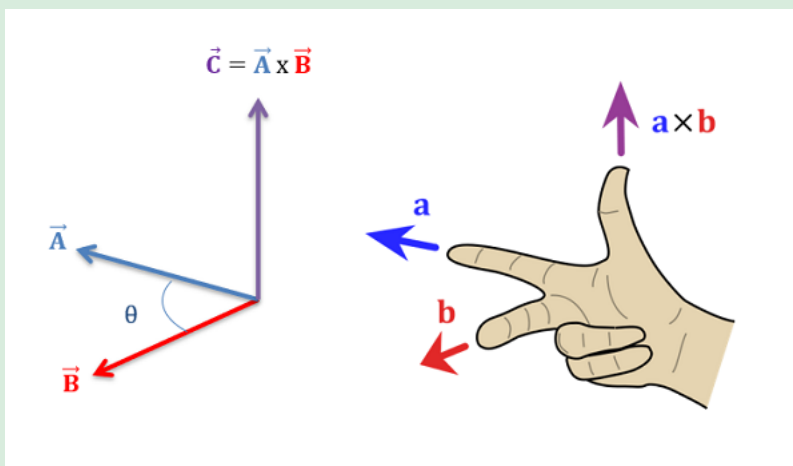
Dot product of two vectors is the length of one of the vectors projected onto the other multiplied by the length of said other vector. Theta = min angle between two vectors.

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$$

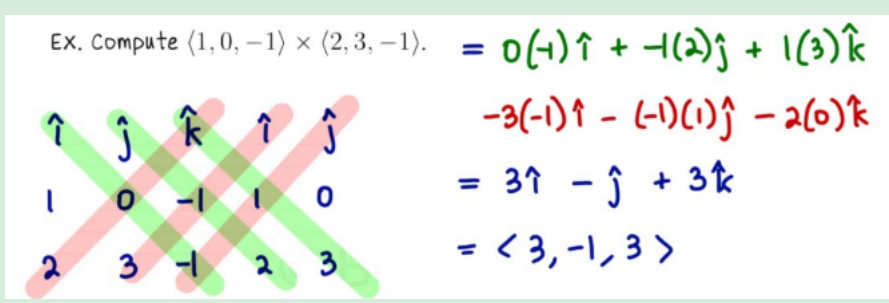
1.1.1 Cross Product

Operation that can be applied to two **three-dimensional** vectors to generate a third vector perpendicular to both vectors. Use the **right hand rule** to deterimine which direction the third vector will go.

Principle 1.1 (Right Hand Rule)
To find the geometrical direction of $\vec{a} \times \vec{b}$, hold your right hand out and point your *index* finger in the direction of $\vec{a}$, and your *middle* in the direction of $\vec{b}$. Your thumb will point in the direction of $\vec{a} \times \vec{b}$.



Principle 1.2 (Calculating the Cross Product)
To calculate the cross product between $\vec{A}$ and $\vec{B}$:



Note the following:

1. $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta$ where theta is the minimum angle betwen the two vectors. Also equal to the area of the parallelogram formed by $\vec{a}$ and $\vec{b}$.
2. $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$. This property is known as the **anti-commutativity of the cross product**.
3. $\vec{a} \times \vec{a} = \vec{0}$ $(\theta = 0)$

1.2 Displacement, Velocity, and Acceleration

$$\vec{v} = \frac{\mathrm{d}\vec{s}}{\mathrm{d}t}, a = \frac{\mathrm{d}\vec{v}}{\mathrm{d}t} = \frac{\mathrm{d}^2\vec{s}}{\mathrm{d}t^2} \implies \Delta\vec{s} = \int \vec{v}(t)dt, \Delta\vec{v} = \int \vec{a}(t)dt$$

Derivatives and intergrals of vector-valued functions can be done component wise.

$$\vec{s} = x(t)\hat{\imath} + y(t)\hat{\jmath} + z(t)\hat{k} \implies \frac{\mathrm{d}\vec{s}}{\mathrm{d}t} = x'(t)\hat{\imath} + y'(t)\hat{\jmath} + z'(t)\hat{k}$$

$$\vec{s} = x(t)\hat{\imath} + y(t)\hat{\jmath} + z(t)\hat{k} \implies \int_{t_i}^{t_f} \vec{s}dt = \int_{t_i}^{t_f} x(t)dt + \int_{t_i}^{t_f} y(t)dt + \int_{t_i}^{t_f} z(t)dt$$

For constant one-dimensional acceleration, kinematic equations apply:

$$\Delta v = \int adt \implies v_f = v_i + at$$

$$\Delta s = \int v(t)dt = \int (v_i + at)dt \implies \Delta s = v_i t + \frac{1}{2}at^2$$

$$v_f = v_i + at \implies v_f^2 = v_i^2 + 2v_i at + a^2t^2 \implies v_f^2 = v_i^2 + a(2v_i t + at^2) \implies v_f^2 = v_i^2 + 2a\Delta x$$

Falling object experience constant acceleration: $a = g \approx 9.81\mathrm{m\,s^{-2}} \approx 10\mathrm{m\,s^{-2}}$. Projectiles experience constant vertical accleration of gravity, but no horizontal acceleration.

1.3 Representing Motion

Distance, velocity, and acceleration vs. time graphs exist. Yipee!

1.4 Reference Frames and Relative Motion

Let $\vec{v}_{AB}$ = velocity of A relative to B.

$$\vec{v}_{AB} = -\vec{v}_{BA}$$

$$\vec{v}_{AC} = \vec{v}_{AB} + \vec{v}_{BC}$$

These are also known as **Galilean Transformations.** Note these equations will break down near the speed of light due to relativity shit.

Example 1.3
The president's airplane flies at $250\mathrm{m\,s^{-1}}$ to the east with respect to the air. The air is moving at $35\mathrm{m\,s^{-1}}$ to the north with respect to the ground. Find the velocity of the president's airplane with respect to the ground.

Answer 1
Let $\hat{\imath}$ = north and $\hat{\jmath}$ = east.

$$\vec{v}_{PG} = \vec{v}_{PA} + \vec{v}_{AG} = (250\hat{\jmath} + 35\hat{\imath})$$

$$|\vec{v}_{PG}| = \sqrt{250^2 + 35^2} \approx 252.4\,\mathrm{m\,s^{-1}}$$

$$\theta_{PG} = \arctan\frac{35}{250} \approx 7.97^\circ \text{ relative to East}$$

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